Finite critical estimates cube partitions and physical cycle obstructions

Consolidated companion proof and reproducibility record

The Eriksson Programme
Preparing assistant: OpenAI Codex; not an independent reviewer

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Abstract

This companion groups the finite critical rectangle estimate, the partition construction for supplied disjoint lattice cubes, and an explicit obstruction to a fixed coarse penalty in the actual flat physical cochain domain. In the last construction kernel coercivity is proved on every actual cochain, while an actual divergence-zero field defeats the same fixed penalty. The complete fixed-penalty statement has been formally checked and reproduced in two fresh Colab clones at one source commit. The earlier Sobolev and cube arguments, the exact subsharp penalty threshold and the sharp free-penalty extension remain analytic; no common verification label is applied to all parts. Suitable cubes are still inputs, not consequences of a critical inverse moment. The unrestricted rectangular question remains open. This is an unreviewed companion revision, not an AIRR deposit, acceptance, peer review or an interacting or continuum mass gap.

Reading convention and provenance

The three parts below have separate scopes. Part one is a local inequality; it is not an algorithm producing a globally adequate partition. Part two constructs a partition only when actual disjoint cubes and the stated coefficient bounds have already been supplied. Part three concerns one explicitly fixed observation map. Their conclusions must not be composed to discard those assumptions.

The dimension symbol D is common to all parts. The constants named A_D are *local to their respective parts*: the Sobolev constant and the geometric partition constant are different. Each is explicitly defined at its point of use; neither is substituted for the other. All lattice norms use counting measure. Internal axis edges are counted once; periodic wrap edges may only be omitted using nonnegativity.

The classical entropy/projection/coarea derivation and de Berg's compressed splitting method are credited in the respective parts. We claim no new general Sobolev or BSP principle, original priority or independent discovery. This revision consolidates source fragments from units137–140, the physical cycle work in units142–148 and the explicit definitions and exact threshold from unit151, with source and existing evidence pinned in the accompanying manifest. It does not amend the frozen reviewed PDF or inherit its model assessments.

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1 A finite proof of the critical rectangle estimate

This supplement derives the balanced-cube endpoint used by the rectangle adapter, rather than importing a local Sobolev estimate as an external input. The method is classical: finite entropy proves the projection inequality, which yields relative edge isoperimetry, layer sets, and a power argument. We claim neither a new general Sobolev principle nor a solution of the open critical rectangular-partition problem. This is part of the existing dossier.

Fix $D \geq 3$, and put

$$p = D/2, \quad q = \frac{2D}{D+2}, \quad q' = \frac{2D}{D-2},$$
$$\alpha = \frac{2(D-1)}{D}, \quad c_D = D(1-2^{-1/D}), \quad K_D = \frac{2\alpha D^{1/q'}}{c_D}.$$

An internal axis edge is an unordered adjacent pair, counted once. For its positive-axis orientation $e = (x, x+e_i)$, write $du(e) = u(x+e_i) - u(x)$.

1.1 Finite projection inequality, with proof

For a nonempty finite $A \subset \prod_{j=1}^{D} X_j$, let π_i delete coordinate i. Then

$$|A|^{D-1} \leq \prod_{i=1}^{D} |\pi_i A|. \tag{1}$$

Here is a finite proof, including the entropy facts needed for this use. For a finite probability vector r, define $H(r) = -\sum r \log r$ with $0 \log 0 = 0$. Concavity follows from $(-t\log t)'' = -1/t < 0$ and continuity at zero. Consequently mixing conditional distributions cannot decrease their average entropy: $H(Y \mid Z) \geq H(Y \mid Z, W)$. Factoring finite joint probabilities proves the chain rule $H(X) = \sum_j H(X_j \mid X_1, \ldots, X_{j-1})$. Also $H(Y) \leq \log |\operatorname{supp} Y|$: Jensen's inequality for log gives $\sum_y r_y \log(1/r_y) \leq \log \sum_y r_y / r_y$ on the positive support. For X uniform on A, $H(X) = \log |A|$, while

$$\begin{aligned}\sum_i H(X_{-i}) &= \sum_i \sum_{j \neq i} H(X_j \mid X_{<j} \setminus X_i) \\ &\geq \sum_i \sum_{j \neq i} H(X_j \mid X_{<j}) = (D-1)H(X).\end{aligned}$$

The notation in the conditioning set means omit X_i only when $i < j$. Bounding each $H(X_{-i})$ by $\log |\pi_i A|$ and exponentiating proves (1). No projection inequality is an unproved hypothesis in the sequel.

1.2 Relative isoperimetry on a finite cube

Let $Q_s = \{0, \ldots, s-1\}^D$, $s \geq 2$, and let $\partial_E A$ consist of internal cube edges crossing A and its complement. If $0 < |A| \leq |Q_s|/2$, then

$$|\partial_E A| \geq c_D |A|^{(D-1)/D}. \tag{2}$$

Indeed, put $a = |A|$, $P_i = |\pi_i A|$. At most a/s occupied fibers in direction i are full. Every occupied nonfull fiber contains an internal crossing edge, because the fiber is a connected interval. Edges in distinct fibers/directions are different. Thus projection inequality and AM–GM give

$$|\partial_E A| \geq \sum_i (P_i - a/s) \geq D a^{(D-1)/D} - Da/s \geq D(1-2^{-1/D}) a^{(D-1)/D}.$$

Empty A satisfies the same inequality trivially. The boundary is the EDGE boundary, not the vertex boundary in a separate graph theorem.

1.3 Endpoint counting-norm Sobolev–Poincare inequality

For every real u on Q_s, including $s = 1$,

$$\sum_{Q_s} |u - \bar{u}_{Q_s}|^2 \le K_D^2 \left(\sum_{e \subset Q_s} |du(e)|^q \right)^{2/q}. \tag{3}$$

For $s = 1$ both sides vanish. For a nonnegative v supported on at most half the cube, layer-set decomposition and the triangle inequality in $\ell^{D/(D-1)}$, followed by (2), imply

$$\|v\|_{D/(D-1)} \le \int_0^\infty |\{v > t\}|^{(D-1)/D}\, dt \le c_D^{-1} \int_0^\infty |\partial_E \{v > t\}|\, dt = c_D^{-1} \sum_e |dv(e)|.$$

The integrals are finite sums of step lengths between values of v; the last equality follows edgewise from the length of the interval between its two endpoint values. Apply this inequality to v^α. For $a, b \ge 0$, the fundamental theorem of calculus gives

$$|a^\alpha - b^\alpha| \le \alpha |a - b| (a^{\alpha-1} + b^{\alpha-1}).$$

Since $\alpha D/(D-1) = 2$ and $(\alpha - 1)q' = 2$, Holder and the degree bound $\deg_{Q_s}(x) \le 2D$ give

$$\begin{aligned} \|v\|_2^\alpha &\le \frac{\alpha}{c_D} \Big(\sum_e |dv(e)|^q \Big)^{1/q} \Big(\sum_{e=\{x,y\}} (v(x)^{\alpha-1} + v(y)^{\alpha-1})^{q'} \Big)^{1/q'} \\ &\le \frac{2\alpha D^{1/q'}}{c_D} \Big(\sum_e |dv(e)|^q \Big)^{1/q} \|v\|_2^{\alpha-1}. \end{aligned}$$

The second inequality uses $(a+b)^{q'} \le 2^{q'-1}(a^{q'} + b^{q'})$ and $\sum_e (v(x)^2 + v(y)^2) \le 2D\|v\|_2^2$. If $v = 0$ the conclusion is immediate; otherwise division proves $\|v\|_2 \le K_D \|dv\|_q$.

Choose a counting median m of u and set $v_+ = (u - m)_+$, $v_- = (m - u)_+$. Both supports have size at most half the cube. For each edge,

$$|dv_+(e)|^q + |dv_-(e)|^q \le |du(e)|^q.$$

For endpoint values on the same side of m this is equality in the nonzero part; on opposite sides it is $a^q + b^q \le (a+b)^q$. Since $2/q \ge 1$, addition of the two squared estimates gives

$$\|u - \bar{u}_{Q_s}\|_2^2 \le \|u - m\|_2^2 \le K_D^2 \big(\|dv_+\|_q^q + \|dv_-\|_q^q \big)^{2/q} \le K_D^2 \|du\|_q^2.$$

This proves (3) without an external graph, ball-adapter, coefficient, or boundary-condition hypothesis.

1.4 Reflection with all aspect-ratio multiplicities

Let $R = \prod_i \{0, \dots, l_i - 1\}$, $V = \prod_i l_i$, $L = \max_i l_i$, and $\kappa = L^2/V^{2/D}$. If $V = 1$ the desired variance bound is zero. Otherwise $L \ge 2$. Put $s = 2L+1$, $r = s^D/V$ and fold Q_s coordinatewise onto R by the endpoint-repeating pattern of period $2l_i$,

$$0, 1, \dots, l_i - 1, l_i - 1, \dots, 1, 0, 0, 1, \dots.$$

For $l_i = 1$ the fold is constant. Since $s/(2l_i) > 1$, each original coordinate value has at least $2\lfloor s/(2l_i) \rfloor \ge s/(2l_i)$ preimages and at most $s/l_i + 2 \le 3s/l_i$ preimages. An original internal edge has at most the same upper count in its own direction; each full period traverses it twice.

Adjacent folded values either coincide or are an original internal edge, possibly with reversed orientation. Thus for the unweighted extension $\widetilde{u}$,

$$\mathrm{Var}_{Q_s}(\widetilde{u}) \geq 2^{-D} r \,\mathrm{Var}_R(u), \qquad \sum_{e\subset Q_s} |d\widetilde{u}(e)|^q \leq 3^D r \sum_{e\subset R} |du(e)|^q.$$

The variance inequality holds by applying the preimage lower bound at the mean of $\widetilde{u}$, and then minimizing over constants on R. Apply (3). Since $2/q - 1 = 2/D$ and $s \leq 3L$,

$$\mathrm{Var}_R(u) \leq A_D \kappa \left(\sum_{e\subset R} |du(e)|^q \right)^{2/q}, \qquad A_D = K_D^2 2^D 3^{2D/q+2}. \tag{4}$$

The factor r is not bounded independently of aspect ratio; it was kept until it became $s^2/V^{2/D}$. Singleton coordinate sides cause no problem.

1.5 Positive original edge weights and field coordinates

For arbitrary positive ORIGINAL edge coefficients b_e, Holder on R with exponents $2/q$ and $2/(2-q)$ gives

$$\left(\sum_e |du|^q \right)^{2/q} \leq \Big(\sum_e b_e |du|^2 \Big) \Big(\sum_e b_e^{-q/(2-q)} \Big)^{(2-q)/q}.$$

Here $q/(2-q) = D/2 = p$. Consequently

$$\mathrm{Var}_R(u) \leq A_D \kappa \Big(\sum_{e\subset R} b_e^{-D/2} \Big)^{2/D} \sum_{e\subset R} b_e |du(e)|^2. \tag{5}$$

No reflected weights were introduced: reflection was completed before this Holder step. For $b_{(x,x+e_i)} = a(x) > 0$, each vertex supplies at most D internal starting edges. Thus the inverse edge mass is at most $D \sum_R a^{-D/2}$, proving exactly the scalar-start estimate

$$\mathrm{Var}_R(u) \leq A_D D^{2/D} \kappa \Big(\sum_R a(x)^{-D/2} \Big)^{2/D} \mathcal{E}_{a,R}(u).$$

For any fixed finite real internal/field dimension, apply this scalar inequality to each coordinate and sum; its coefficient factor is common. There is no additional internal-dimension factor and no replacement of original edges by enlarged-region energy.

1.6 Necessity of geometry and status

For uniform $a > 0$ and a longest coordinate of length $L \geq 2$, put $u(x) = x_i - (L-1)/2$. Direct sums give

$$\mathrm{Var}_R(u) = V(L^2-1)/12, \qquad \mathcal{E}_{a,R}(u) = aV(L-1)/L,$$

so their ratio is $L(L+1)/(12a) \geq L^2/(12a)$. Since $\kappa(\sum_R a^{-D/2})^{2/D} = L^2/a$, the geometric dependence cannot be uniformly dropped even for uniform weights. This is not a global counterexample to a partition algorithm or to the physical divergence-zero observation problem.

For attribution and comparison see S. Andres, J.-D. Deuschel and M. Slowik, *Harnack inequalities on weighted graphs and some applications to the random conductance model*, arXiv:1312.5473v6, Section 2.2, (2.5)–(2.8). Their local normalized inequalities are classical prior context; this proof does not import their theorem, graph assumptions or Harnack conclusions. The finite projection/entropy, coarea and power steps above are classical, not a new general method. The gain is a complete finite derivation of the specific input previously left external. No Lean execution, independent review, original-priority claim, new physical hypothesis removal or new AIRR paper follows from this supplement. The frozen PDF candidate and its actual reports/permissions remain unchanged; integration would create a new exact artifact and must not inherit their assessments.

2 A finite rectangular partition of disjoint lattice cubes

The linear partition for supplied disjoint cubes can be derived directly, without importing a general fat-object or low-density theorem. We use the classical compressed cube-splitting strategy of de Berg, UU-CS-1998-12, Section 2.1 and Lemma 2.1, and prove the required special-case geometry and all discrete/energy adapters below. This is not an original general BSP method. No time-complexity, measured implementation or Lean claim is made.

Fix $D \geq 3$, an integer $n \geq 1$, and pairwise disjoint nonempty integer cubes $S_k = \prod_i \{b_{ki}, \ldots, b_{ki} + L_k - 1\}$ in the common root $X = \{0, \ldots, n-1\}^D$, for $1 \leq k \leq J$. Set

$$Q_k = \prod_i [b_{ki} - 1/4, b_{ki} + L_k - 3/4].$$

These are positive-side closed solid cubes, including $L_k = 1$. They are strictly disjoint: disjoint integer intervals in a separating coordinate leave a real gap at least $1/2$. Their integer points are exactly S_k, and each such point is interior to Q_k. All corners of the Q_k lie strictly inside $[-1, n]^D$.

2.1 Proper-intersection corner bound

Let C be any axis-aligned cube of side $s > 0$ with no corner of any Q_k in its interior. Then at most 2^D of the Q_k meet $\operatorname{int} C$. Indeed, if Q_k of side $a < s$ properly meets C, each of its coordinate intervals has an endpoint in the interior of the corresponding interval of C. Choosing these endpoints gives a corner of Q_k in $\operatorname{int} C$, a contradiction. Thus every properly meeting Q_k has $a \geq s$. In each coordinate its interval then contains at least one endpoint of the interval of C; otherwise it would lie strictly inside that shorter interval. Hence Q_k contains a corner of C. Strict disjointness makes these corner choices injective, proving the bound.

The qualifier "properly" is essential. Closed boundary-only contact must not be counted here. For $C = [0, 1]^D$ and any integer $K \geq 1$, put $\epsilon = 1/(4K + 4)$ and

$$Q_j = [-\epsilon, 0] \times [(2j+1)\epsilon, (2j+2)\epsilon] \times [\epsilon, 2\epsilon]^{D-2}, \qquad 0 \leq j < K.$$

These K strictly disjoint small cubes touch C but none meets its interior or has a corner in its interior. Thus the corresponding closed-contact bound would be false. This boundary guard does not assert that de Berg's theorem is false; our construction below explicitly uses open-cell intersections and interior lattice points.

2.2 A multiple-object stress family

The corner-free property does not imply that a query cube meets only one input cube. This can fail even for the quarter-offset solids of equal-side integer cubes, not merely for arbitrary real objects. For $D \geq 2$, put $n = 4$, $C = [1, 2]^D$, and, for $\sigma \in \{0, 1\}^{D-1}$, set

$$b_\sigma = (2\sigma_1, \ldots, 2\sigma_{D-1}, 1), \quad L_\sigma = 2, \qquad S_\sigma = \prod_{i=1}^{D} \{b_{\sigma i}, b_{\sigma i} + 1\}.$$

The 2^{D-1} integer cubes are pairwise disjoint: distinct binary words have a coordinate with the disjoint integer intervals $\{0, 1\}$ and $\{2, 3\}$. Their solids have, in that coordinate, intervals $[-1/4, 5/4]$ and $[7/4, 13/4]$, with a gap $1/2$. In the last coordinate every solid has interval $[3/4, 9/4]$. Consequently every solid properly meets C, while no solid corner is interior to C: every such corner has last coordinate $3/4$ or $9/4$. Thus a universal proper-occupancy bound for corner-free query cubes in this exact lattice class must be at least 2^{D-1}. This neither proves sharpness of 2^D nor asserts that this particular C occurs as a leaf of our root-based recursion. It rules out replacing the occupancy argument by a one-object assumption. The finite unit139 baseline leaves all had occupancy at most one, so those runs did not test this phenomenon.

2.3 Constructing linearly many vertex-free rectangles

Let V be the set of all corners of the Q_k, so $N = |V| = 2^D J$. For $J = 0$ use the single root rectangle and skip this construction. For $J > 0$ begin with the cube $[-1, n]^D$. At a current cube C keep exactly $V_C = V \cap \operatorname{int} C$. If V_C is empty, stop at that cube. Otherwise consider its 2^D open half-side children.

If the points are not all in one open child, perform all midpoint cuts and recurse on the nonempty child point sets. Points on the cuts are removed from subsequent interior sets. If all points lie in one open child, reflect and translate coordinates so the common corner is zero and $C = [0, s]^D$. Put

$$a = \max\{x_i : x \in V_C,\ 1 \le i \le D\}, \qquad C' = [0, a]^D.$$

Then $0 < a < s/2$. Successively cut at the D upper facet planes of C', producing C' and the D boxes

$$E_i = [0, a]^{i-1} \times [a, s] \times [0, s]^{D-i}, \qquad 1 \le i \le D.$$

Use disjoint boundary ownership for these boxes. Only C' can contain points of V in its interior, and at least one point of V_C lies on its boundary. Stop at the other boxes and recurse inside C' if needed. Every intermediate or terminal cell is full-dimensional; a boundary ownership convention never creates a separate hyperplane cell.

Here is a direct finite termination and size proof. Among active cells, let $n_C = |V_C| > 0$ and use the nonnegative integer potential

$$\Psi = \sum_{C \text{ active}} (2n_C - 1).$$

Initially $\Psi = 2N - 1$. If a processed cell loses r points to new cut boundaries and has t nonempty child point sets, the decrease is $2r + t - 1$. In the shrinking step $r \ge 1$. In a midpoint step either $r \ge 1$ or $t \ge 2$. Thus every event decreases Ψ by at least one, so there are at most $2N - 1$ events. Each event creates at most $2^D - 1$ additional leaves, since a shrinking step creates only D extra leaves. The number M of terminal rectangles satisfies

$$M \le 1 + (2N - 1)(2^D - 1) \le 2^{D+1}(2^D - 1)J. \tag{6}$$

Tracking exactly $V \cap \operatorname{int} C$ ensures that every terminal rectangle has no corner of an input cube in its interior. This argument is a construction and a decreasing integer potential, not a supplied partition certificate. It does not prove a fast implementation bound.

2.4 A uniform occupancy bound for the terminal rectangles

Terminal cubes meet at most 2^D input cubes properly, by the corner bound. A terminal noncube has the form E_i above after a coordinate reflection. It is contained in the slab $[0, s]^{i-1} \times [a, s] \times [0, s]^{D-i}$. Cover this slab by 2^{D-1} closed cubes of side $s - a$: in coordinate i use $[a, s]$, and in every other coordinate choose either $[0, s - a]$ or $[a, s]$. These intervals cover $[0, s]$ since $a < s/2$. Every covering cube lies inside the parent cube and outside the interior of C', and its interior contains no point of V. The corner bound therefore applies separately to each covering cube.

If an input cube properly meets E_i, the intersection contains an open set. A point of that set avoiding the finitely many covering-facet planes belongs to the interior of a covering cube. Thus all proper intersections are counted by the cover, even if some particular point lies on a cover boundary. Consequently each terminal rectangle properly meets at most

$$T_D = 2^{D-1} 2^D = 2^{2D-1} \tag{7}$$

input cubes. The covering argument, including its boundary qualification, is proved here; no arbitrary-box unclutteredness is assumed.

2.5 From open geometry to an actual integer partition

For each terminal rectangle list the at most T_D cubes meeting its interior. Refine by all their facet coordinates. There are at most $2T_D$ cut coordinates per axis, so at most $(2T_D+1)^D$ refined rectangles per terminal cell. All cuts are axis-parallel. Now assign every splitting hyperplane to one child consistently, intersect with X, and delete empty integer sets. The result is a disjoint cover of X by integer rectangles. Taking integer points and deleting empty sets does not increase the count. More explicitly, use the uniform convention $[l_i, h_i)$ in every axis of every cell. A coordinate split at t then assigns the plane to the upper child. The two intervals $[l,t)$ and $[t,h)$ partition $[l,h)$ exactly, even when t is an integer. By induction through every cut, these half-open cells partition $[-1,n)^D$; intersecting with X gives

$$R = \prod_i (\{\lceil l_i \rceil, \dots, \lceil h_i \rceil - 1\} \cap \{0, \dots, n-1\}).$$

This formula is also valid for negative or integer endpoints. An empty axis makes the entire integer cell empty; no lattice point is assigned twice or lost when those cells are deleted. The outer upper face at n contains no point of X.

To check membership rigorously, suppose an assigned integer point x lies in S_k. It is interior to Q_k, while its assigned terminal rectangle has x in its closure. A small ball about x lies in Q_k and meets the interior of that full-dimensional terminal rectangle. Therefore Q_k was in its list. None of the new facet coordinates is an integer. Hence each refined integer rectangle is entirely inside a listed S_k or disjoint from every listed S_k. An unlisted cluster cannot contain one of its integer points, by the same closure argument. Strict disjointness then implies that each integer rectangle is inside one cluster or outside every cluster.

In particular, with the explicit dimension-only constant

$$A_D = 2^{D+1}(2^D - 1)(2^{2D} + 1)^D, \qquad |\mathcal{P}| \le A_D \max\{1, J\}, \tag{8}$$

we have derived the required partition. The constant is deliberately loose and no sharpness or algorithm running-time claim is made. For $n = 1$ and for singleton input cubes all constructions remain valid.

2.6 The actual coefficient and fixed-penalty consequence

Suppose, in addition, that for fixed $c > 0$ and $\beta \ge 1$ the actual positive scalar starting coefficients obey

$$\begin{aligned} a(x) &\ge cL_k(L_k - 1)/4 \quad (x \in S_k), \\ a(x) &\ge cn(n-1)/4 \quad (x \notin \cup_k S_k), \\ a(x) &\le \beta L_k^2 \quad (x \in S_k). \end{aligned}$$

These coefficient/cube assumptions are retained, not inferred from the critical moment. On an interval of l integer vertices, pairwise variance, path Cauchy–Schwarz and at most $l^2/4$ pairs through one cut give

$$\sum |u - \bar{u}|^2 \le \frac{l(l-1)}{4} \sum_{\text{edges}} |du|^2.$$

For completeness, product variance tensorization follows from the commuting coordinate averages $P_i u = l_i^{-1} \sum_{t=0}^{l_i - 1} u(x_1, \dots, t, \dots, x_D)$: the telescoping orthogonal decomposition of $I - \prod_i P_i$ gives a sum bounded by $\sum_i \|u - P_i u\|_2^2$. Applying the interval bound on its fibers proves

$$\mathrm{Var}_R(u) \le \frac{L(R)(L(R)-1)}{4} \sum_{e \subset R} |du(e)|^2.$$

Inside a cluster $L(R) \le L_k$, and outside all clusters $L(R) \le n$. The displayed coefficient lower bounds therefore give $c\,\mathrm{Var}_R(u) \le \mathcal{E}_{a,R}(u)$. Singleton residuals are zero, so no zero lower bound is divided by. Sum over all field/internal coordinates and all partition cells. Original internal positive-axis edge occurrences are disjoint; omitted inter-cell and periodic-wrap occurrences have nonnegative energy. Thus the ACTUAL normalized means

$$(C_{\mathcal{P}}U)_{R,j} = \sqrt{|R|}\,\mathrm{mean}_R\, U_j$$

satisfy

$$c\|U\|^2 \le \mathcal{E}_a(U) + c\|C_{\mathcal{P}}U\|^2$$

for every cochain, with output dimension $h|\mathcal{P}|$, $h = D(N_c^2 - 1)$. This uses an all-cochain variance estimate, not the invalid inference from a kernel-only margin to fixed penalty c. Means need not preserve the divergence constraint. The inequality consequently also holds on that constraint subspace without changing $C_{\mathcal{P}}$.

If $p = D/2$, $\mu \ge 1$, and the original critical mass bound is $\sum_X a^{-p} \le Mn^D/\mu^p$, then each supplied cube contributes at least $L_k^D/(\beta L_k^2)^p = \beta^{-p}$ to this mass. Hence $J \le \beta^p Mn^D/\mu^p$, and (8) gives

$$\dim F \le hA_D(1 + 2^p\beta^p M)\max\{1, n^D/(1+\mu)^p\}.$$

This proves the restricted supplied-cube consequence with an explicit constant and without a fat-object BSP theorem as a black box. It does not construct the supplied cubes from the critical moment, remove either scale bound or the strong complement bound, resolve the unrestricted critical rectangular question, or establish a covariant/continuum gap. No new originality or independent-discovery claim is made: the compressed splitting strategy is attributed classical geometry, here derived and adapted completely for this special class. The frozen reviewed PDF and its full reports and pending permission remain unchanged.

3 Physical cycle obstructions for an actual observation map

This part closes a limitation of the scalar four-cycle example: the obstruction is now realized inside the actual divergence-zero flat physical cochain domain. The kernel estimate below is proved for all actual cochains, not just the special fields used as negative witnesses. The conclusion concerns this observation and this fixed penalty. It does not obstruct all partitions or the optimal-rank question from the preceding parts.

3.1 The actual form and rectangular observations

Fix $D \geq 3$, $N_c \geq 2$ and $X = (\mathbb{Z}/4\mathbb{Z})^D$. Write $V = \mathbb{R}^{N_c^2-1}$ with its standard Euclidean norm and $A = (A_\ell(x))_{x\in X, 1\leq \ell\leq D}$ for an actual one-cochain. All norms use counting measure. Choose the actual total coefficients

$$\alpha = (1, 6, 4, 1), \qquad a(x) = \alpha_{x_1}, \qquad w(i, x, \ell) = \alpha_{x_1} - 1$$

for every derivative direction i and physical component ℓ. Define actual forward and backward differences by

$$D_i^+ f(x) = f(x + e_i) - f(x), \qquad D_i^- f(x) = f(x) - f(x - e_i).$$

The flat curl and backward divergence, with one curl for each unordered pair, are

$$(\operatorname{curl} A)_{ij} = D_i^+ A_j - D_j^+ A_i \quad (i < j), \qquad \operatorname{div}^- A = \sum_i D_i^- A_i.$$

Their counting quadratic form and the original componentwise repair are

$$H_0(A) = \sum_{i<j,x} \|D_i^+ A_j(x) - D_j^+ A_i(x)\|^2 + \sum_x \Big\| \sum_i D_i^- A_i(x) \Big\|^2,$$
$$R_w(A) = \sum_{i,x,\ell} w(i, x, \ell) \|D_i^+ A_\ell(x)\|^2.$$

In operator notation the repair is $\langle A, G^* W G A\rangle$ for the actual stacked gradient G and diagonal multiplication W. These formulas fix the energy on paper; no comparison certificate is supplied.

Deriving the flat identity. Periodic reindexing gives

$$\sum_x \langle D_i^+ f, h\rangle = -\sum_x \langle f, D_i^- h\rangle.$$

Coordinate shifts commute. Applying this formula twice and real symmetry therefore yields

$$\sum_x \langle D_i^+ f, D_j^+ g\rangle = \sum_x \langle D_j^- f, D_i^- g\rangle.$$

Also $\sum_x \|D_i^- f\|^2 = \sum_x \|D_i^+ f\|^2$ by translation. Expand each curl square with $f = A_j$, $g = A_i$: it supplies the two off-diagonal gradient norms and the negative mixed terms. Expansion of the divergence square supplies the diagonal norms and the opposite mixed terms. Cancellation for each $i < j$ proves, for every cochain,

$$H_0(A) = \sum_{i,x,\ell} \|D_i^+ A_\ell(x)\|^2.$$

No divergence condition or lift ansatz was used. Define $H = H_0 + R_w$. Adding the displayed repair, with $1 + w(i, x, \ell) = \alpha_{x_1}$, gives

$$H(A) = \sum_{x \in X} \sum_{i=1}^{D} \sum_{\ell=1}^{D} \alpha_{x_1} \|A_\ell(x + e_i) - A_\ell(x)\|^2. \tag{9}$$

The cancellation was unweighted; the varying repair is added afterwards. We do not move α through summation by parts or assert a weighted curl–divergence identity. This is the paper counterpart of `finite_flat_curl_div_energy` and `componentwiseWeightedRepair_energy` in the pinned formal source. Thus the coefficient is the original shared scalar total coefficient, not a substituted energy or an interacting Hessian.

Put $I_0 = \{0, 1\}$ and $I_1 = \{2, 3\}$. For every transverse $y \in (\mathbb{Z}/4\mathbb{Z})^{D-1}$ the blocks $I_b \times \{y\}$ are nonempty axis-aligned rectangles. They are disjoint and cover X. For each physical component ℓ and each standard internal coordinate k, define the actual normalized means

$$(CA)_{b,y,\ell,k} = \frac{1}{\sqrt{2}} \sum_{t \in I_b} A_\ell(t, y)_k. \tag{10}$$

The output includes *all* these coordinates. If $h = D(N_c^2 - 1)$, its dimension is $2h4^{D-1}$; no optimality is asserted.

The spatial mean of a is $\mu = 3$, and $1 \le a \le 2\mu$. For $p = D/2$ its original normalized inverse moment is

$$\frac{1}{|X|} \sum_{x \in X} \left(\frac{\mu}{a(x)} \right)^p = \frac{2 \cdot 3^p + (1/2)^p + (3/4)^p}{4} =: M_D < \infty. \tag{11}$$

This is an actual member of the stated bounded-moment coefficient class when the allowed bound is at least M_D. Its fixed finite size and the large output dimension do not disprove an asymptotic optimal-rank order.

3.2 A fixed-penalty counterexample with a proved kernel premise

For a real cyclic vector $u = (u_0, u_1, u_2, u_3)$ set

$$E(u) = (u_1 - u_0)^2 + 6(u_2 - u_1)^2 + 4(u_3 - u_2)^2 + (u_0 - u_3)^2,$$
$$|u|^2 = \sum_{t=0}^{3} u_t^2, \qquad O(u) = \frac{(u_0 + u_1)^2 + (u_2 + u_3)^2}{2}.$$

Choose a unit internal coordinate vector T. The actual lift is

$$(Ju)_2(x) = u_{x_1} T, \qquad (Ju)_\ell(x) = 0 \quad (\ell \neq 2).$$

Its backward divergence is zero: its only nonzero component is constant in coordinate 2. It is injective and direct summation gives

$$\|Ju\|^2 = 4^{D-1}|u|^2, \quad H(Ju) = 4^{D-1}E(u), \quad \|CJu\|^2 = 4^{D-1}O(u). \tag{12}$$

The last equality uses the entire output in (10), not only a selected nonzero coordinate.

Kernel premise on all actual fields. For $v = (x, -x, y, -y)$, direct algebra yields

$$E(v) = 11x^2 + 14xy + 23y^2, \qquad |v|^2 = 2x^2 + 2y^2,$$
$$E(v) - \frac{19}{5}|v|^2 = \frac{17}{5}\left(x + \frac{35}{17}y\right)^2 + \frac{84}{85}y^2 \geq 0. \tag{13}$$

If $CA = 0$, both pair sums vanish for every actual scalar profile $t \mapsto A_\ell(t, y)_k$. Apply (13) to each such profile and sum over all y, ℓ, k. These profiles account for the full norm of A. Their energies are exactly the $i = 1$ term of (9); every other derivative term is nonnegative because $\alpha_t \geq 1$. Therefore

$$CA = 0 \quad \Longrightarrow \quad H(A) \geq \frac{19}{5}\|A\|^2 \qquad \text{for every actual cochain } A. \tag{14}$$

No divergence condition, scalar-lift ansatz or supplied coercivity certificate was used in this proof.

Failure on an actual divergence-zero field. For $u = (19, -9, -5, -5)/14$ one has

$$E(u) = \frac{52}{7}, \qquad |u|^2 = \frac{123}{49}, \qquad O(u) = \frac{25}{49}.$$

Consequently the actual divergence-zero field Ju satisfies

$$H(Ju) + \frac{19}{5}\|CJu\|^2 - \frac{19}{5}\|Ju\|^2 = -\frac{6}{35}4^{D-1} < 0. \tag{15}$$

Equations (14) and (15) prove the complete fixed-map statement: kernel coercivity does not imply the same coefficient as a fixed coarse penalty, even on the actual flat physical divergence-zero domain. They do not rule out larger penalties or different observations.

3.3 The exact finite penalty at the subsharp target

The failure just proved concerns $K = 19/5$. Here we determine exactly which larger penalties work for this same observation and target. This subsection is a paper proof with an exact rational coefficient check; it is not part of the paired Lean theorem below.

Every $u \in \mathbb{R}^4$ has the unique decomposition

$$u = m(1, 1, 1, 1) + (x, -x, y, -y) + (\delta/2, \delta/2, -\delta/2, -\delta/2),$$

where $m = (\sum_t u_t)/4$, $x = (u_0 - u_1)/2$, $y = (u_2 - u_3)/2$ and $\delta = (u_0 + u_1 - u_2 - u_3)/2$. Direct expansion gives

$$|u|^2 = 4m^2 + 2x^2 + 2y^2 + \delta^2, \qquad O(u) = 4m^2 + \delta^2,$$
$$E(u) = 11x^2 + 14xy + 23y^2 + 7\delta^2 - 10\delta(x + y).$$

Set $X = x - 25\delta/2$ and $Y = y + 75\delta/14$. For every real K,

$$E(u) + KO(u) - \frac{19}{5}|u|^2 = 4\left(K - \frac{19}{5}\right)m^2 + \frac{17}{5}\left(X + \frac{35}{17}Y\right)^2 + \frac{84}{85}Y^2 + \left(K - \frac{1138}{35}\right)\delta^2. \tag{16}$$

Since $1138/35 > 19/5$, this proves sufficiency of $K \geq 1138/35$ on every scalar profile. Necessity follows from

$$u_* = (13, -12, -41/7, 34/7), \qquad O(u_*) = 1, \quad |u_*|^2 = \frac{18174}{49}, \quad E(u_*) = \frac{67468}{49}.$$

It has $m = 0$, $\delta = 1$ and $X = Y = 0$, hence defect $K - 1138/35$.

Apply the scalar inequality to every profile $t \mapsto A_\ell(t, y)_k$ and sum over all y, ℓ, k. Their squared norms and observations exhaust the actual full norms, and their energies exhaust the first-direction contribution in (9). The remaining directions are nonnegative. Conversely Ju_* is actual divergence zero and multiplies the scalar defect by 4^{D-1}. Therefore the following are equivalent:

1. $K \geq 1138/35$;
2. $H(A) + K\|CA\|^2 \geq (19/5)\|A\|^2$ for every actual cochain;
3. the same inequality for every actual divergence-zero cochain.

This is a necessity-and-sufficiency statement for the fixed map, not an obstruction to different partitions. In particular, a larger finite penalty repairs the subsharp target even though $K = 19/5$ fails. The sharp kernel target in the next subsection behaves differently.

A second calculation records the classical block elimination behind the constant. The kernel residual matrix and mixed vector are

$$M = \begin{pmatrix} 17/5 & 7 \\ 7 & 77/5 \end{pmatrix}, \quad \det M = 84/25, \quad b = (5, 5), \quad M^{-1}b = (25/2, -75/14).$$

Thus $b^T M^{-1} b = 250/7$ and the residual coefficient is $K + 16/5 - 250/7 = K - 1138/35$, agreeing with (16). No new Schur-complement principle is claimed.

The unit151 exact verifier constructs E, O and $|u|^2$ from the actual four-coordinate linear forms and compares every coefficient of the polynomial in (m, x, y, δ, K), rather than sampling values. Normal and optimized Python in one own Colab Linux runtime each accepted five checks and rejected three real coefficient mutations; both exits were zero and their JSON outputs were identical. The 8794-byte archive has ten members and nine checked inner hashes. It and the full hand derivation are included in the reproduction package. The driver took about 0.56 seconds; the runtime was released after recovery and exact billed consumption is unknown. This check is neither a new Lean theorem nor independently authored verification. Its source is

`564b84230478cb8d40505cb67985470e9bae699d.`

3.4 Sharp free-penalty extension and its distinct status

The following is an analytic continuation, outside the paired Lean result. Put $s = \sqrt{85}$ and $c_* = (17 - s)/2$. Since $9 < s < 10$, the exact identity

$$E(v) - c_*|v|^2 = (s - 6)\left(x + \frac{7y}{s - 6}\right)^2$$

holds on the same block kernel. The nonzero vector $v_* = (7, -7, 6 - s, s - 6)$ attains equality. The all-profile argument above also establishes the sharp kernel constant for the actual form in paper proof, and the lift establishes sharpness.

For $p_\delta = (\delta/2, \delta/2, -\delta/2, -\delta/2)$, the two vectors are orthogonal, $Cv_* = 0$, and $O(p_\delta) = |p_\delta|^2 = \delta^2$. For any finite $K \geq 0$, direct expansion gives

$$\begin{aligned} E(v_* + p_\delta) + KO(v_* + p_\delta) - c_*|v_* + p_\delta|^2 \\ = (7 + K - c_*)\delta^2 - 10(13 - s)\delta. \end{aligned}$$

The positive number $q_K = 7 + K - c_*$ permits the choice $\delta = 5(13 - s)/q_K$, giving the strictly negative value $-25(13 - s)^2/q_K$. Lifting multiplies this complete defect by 4^{D-1} and preserves

actual divergence zero. Thus no finite quadratic penalty repairs this same observation at the sharp kernel endpoint. A penalty $\Phi(z) \le K\|z\|^2$ only near zero is also defeated: scale the negative witness by a sufficiently small nonzero real number. Both radius conditions can be met, and its defect remains a negative multiple of the square of that scale. This is not a strict-margin claim.

These exact analytic identities and arguments are supplied here, but their complete physical sharp-endpoint/free-penalty statement has *not* been verified in Lean in the paired execution described next. Finite checks in dimensions three and four, including $K = 0, 1, 100$, are diagnostics only, not the universal theorem or independent review.

3.5 Formal scope and the paired reproduction record

The fixed-penalty result, including the actual full-kernel premise and all transport/comparison identities used to connect it, was compiled in two separately allocated Colab Linux CPU sessions with high RAM. Each started with a fresh clone at the same immutable source

```
0c706827dcbb11859e56bb6c7dc981eb37a02e69.
```

The final theorem is named `physicalCycleFour_kernel_fixed_penalty_counterexample` in `PhysicalCycleFullForm.lean`. It asserts the proved all-cochain kernel bound together with failure of the fixed-penalty inequality on the actual divergence-zero domain. Four preceding printed endpoints identify longitudinal energy, the full actual energy sum, its comparison, and the kernel estimate. They do not assume transport certificates or posit a structure containing the conclusion.

All five actual dependency readouts in each clone contain only `propext`, `Classical.choice` and `Quot.sound`. The normal and optimized validator in each clone accepted all five readouts and rejected three actual oracle mutations. All actual decimal exit sentinels were zero. Lean was version 4.29.0-rc6 with Mathlib pinned in the included manifest. The recovered theorem source, oracle source, compiled binary, oracle output, validator source, normal and optimized outputs and driver have identical hashes across the two fresh executions. The raw per-run reports still say first-clone pass and paired false: the identical driver executes one clone per session. They remain unmodified; the separate aggregate records the completed pair.

The first successful driver took about 374.31 seconds and the second 400.28 seconds. These are not billed connection times. Both environments were disconnected after complete evidence recovery. No Lean compilation, oracle or numerical campaign ran on Windows. Separate fresh executions are not independently authored proofs or independent scientific review. The previous failed attempt, its exact source and full compiler log are preserved: mixed finite-index inference and nested sum type inference were corrected without changing the theorem statements.

The package includes both public evidence archives. Their comparison is recorded in the aggregate file:

```
paired-full-form148.json
```

Full hashes and source bindings are in `inputs.json`. Each archive has 45 checked inner manifest members and excludes third-party installers and downloaded binary assets. Their complete originals, including those assets, are retained in the private evidence record. The compiled binary hash begins `b0f8ff8654d79d02`, and the full value is in the paired comparison file.

Attribution and scope of novelty. The scalar square identities and negative vectors predate this physical embedding in the dossier. The present consolidation adds the explicit actual-domain embedding and the formally derived all-field comparison; it claims no new

Schur-complement principle, priority or exhaustive novelty finding. Carlson, Haynsworth and Markham's classical paper, *A Generalization of the Schur Complement by Means of the Moore–Penrose Inverse*, SIAM Journal on Applied Mathematics 26 (1974), 169–175, doi:10.1137/0126013, supplies algebraic context. The publisher abstract and bibliography were inspected in the source investigation; the full paywalled article was not read. The direct finite proof here carries the mathematical claim.

4 Finite reproduction evidence and its limits

The construction was implemented using exact rational arithmetic. On 5 November 2026 a newly created, exclusively owned Colab Linux CPU runtime with high RAM ran source commit

```
3c309786dff766a97638b9bb4f654bcb14cb9455
```

under Python 3.13.18. Normal and optimized Python each accepted eight baseline certificates and rejected all 47 saved malformed certificates and two invalid inputs. All 58 canonical JSON files matched byte for byte between the two modes. The source had zero assertion nodes; acceptance used explicit failure checks. Both subprocess exit codes were zero. Seven scenes used exhaustive lattice coverage. The scene with root side 10^9 used exact volume equality and pairwise disjointness instead. The checker reconstructed split geometry, active corner sets and potential decrease, and checked whole-rectangle membership and coverage. Midpoint and corner-shrink events both occurred. Full logs, real exit sentinels, actual malformed certificates and sources are retained in the archive.

This checker shares primitives and authorship with the builder. It is not an independently authored validator or reproduction in two independent fresh clones. These are finite fixture results, not a proof certificate for the universal theorem or Lean verification. All baseline first leaves had proper occupancy at most one. They therefore did not test the multiple-object family proved in the preceding part. Closed-contact negative controls in dimensions two, three and four had respectively five, nine and seventeen closed contacts but no proper contact.

The archive is 193158 bytes. After recovery from Colab all 123 inner manifest members were rehashed without a mismatch. Its SHA256 is

```
64f548dff043cf6cf9f1c1a3abf24fc79
3cf6565ff7dd436c1d0f263351b5be1
```

The line break in this printed hash is typographical; concatenate the two lines. The full archive is included in the reproduction package. The runtime was disconnected after recovery. The driver took approximately 3.88 seconds; this is not the billed connection time or an algorithmic complexity bound. The resource panel displayed 50.99 GB RAM; an exact billed compute-unit delta was unavailable and is not inferred.

5 Model assessments and review context

This exact successor companion PDF has no external model assessment and no peer review. Preparing-assistant analytic checking and visual layout checking are reported separately from external review. The authoring interface is OpenAI Codex; no more specific API-attested model identifier is available in this record. No independence from the proof's preparing context is claimed. No review score is assigned to this artifact.

For historical context only, the distinct frozen manuscript identified by the SHA256 prefix `eb4df3` has its own complete first model report and review context retained in revision136. The original report gives 3.5/10 and minor revision; the raw model identifier is unknown and the operator observed Google Gemini 4 Argon High in the interface. Its five material objections concern uninspected formal/Colab evidence, the non-Lean analytic continuation, primary sources not inspected by that reviewer, the still-open unrestricted rectangular question, and the absent interacting/continuum connection. These remain part of that artifact's report. They have not been erased or reassessed by this companion. That review was not blinded; prior assessment narrative was in its PDF. Independence and context isolation were not established.

For the distinct preceding 14-page companion, SHA256 prefix `5c2774`, the first external model report is retained in full in `model-assessment150`. It recommends minor revision and assigns an AI-only 6.5/10. The operator selected Google Gemini 4 Argon High; the served backend identifier is unexposed and the report itself says unknown. One request was made, without a target grade or regeneration. The PDF's historical narrative made that review non-blinded. Neither independent authorship nor context isolation is established.

All five objection groups remain available: absent self-contained Hodge/repair definitions and uninspected execution archives; the gap between a local critical inequality and supplied cubes; the distinction between fixed, subsharp and sharp penalties; alleged technical defects; and uninspected primary sources and non-blinded context. The four technical allegations were contradicted by the exact previous PDF: $a \geq s$ and $a, b \geq 0$ were already printed, $2\,3^p$ was a product, and $O(p_\delta) = |p_\delta|^2 = \delta^2$ was complete over a line break. The original report and its 6.5/10 are unchanged; a separate operator note records the evidence and the report's inconsistent score subtotals. The present version adds the explicit energy definitions and cancellation and the exact finite penalty threshold. It does not make that reviewer inspect archives, primary sources or this new PDF. All remaining scope limitations stay in force, and no score transfers to this artifact. The public AIRR v5 is a separate artifact, a working paper rather than an accepted or peer-reviewed paper. Its own first report and operator response remain separate. No historical rating transfers to this new companion. Public deposit, model evaluation, editorial acceptance and peer review are different events; no new such event is claimed here.

6 Unresolved obligations and reproducibility instructions

The unrestricted critical rectangular partition question is open. The generic common-refinement mechanism remains stopped after its recorded counterexamples; these classical proof completions do not restart it. All actual cube, coefficient lower/upper, strong complement and moment assumptions in part two remain. No interacting physical hypothesis is removed. PR29 remains recorded as 20/41 with TermSource zero.

Compile this main file with the pinned fragments at the manifest source SHA. Retain every LaTeX log, exact PDF hash and page image. Rendering runs in an own Colab Linux runtime and does not rerun the saved fixtures or formal pair. Compilation does not establish mathematics. Frozen reviewed PDFs and reports remain unchanged.